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Introduction to cognitive science

Homework 2 – Phase 2

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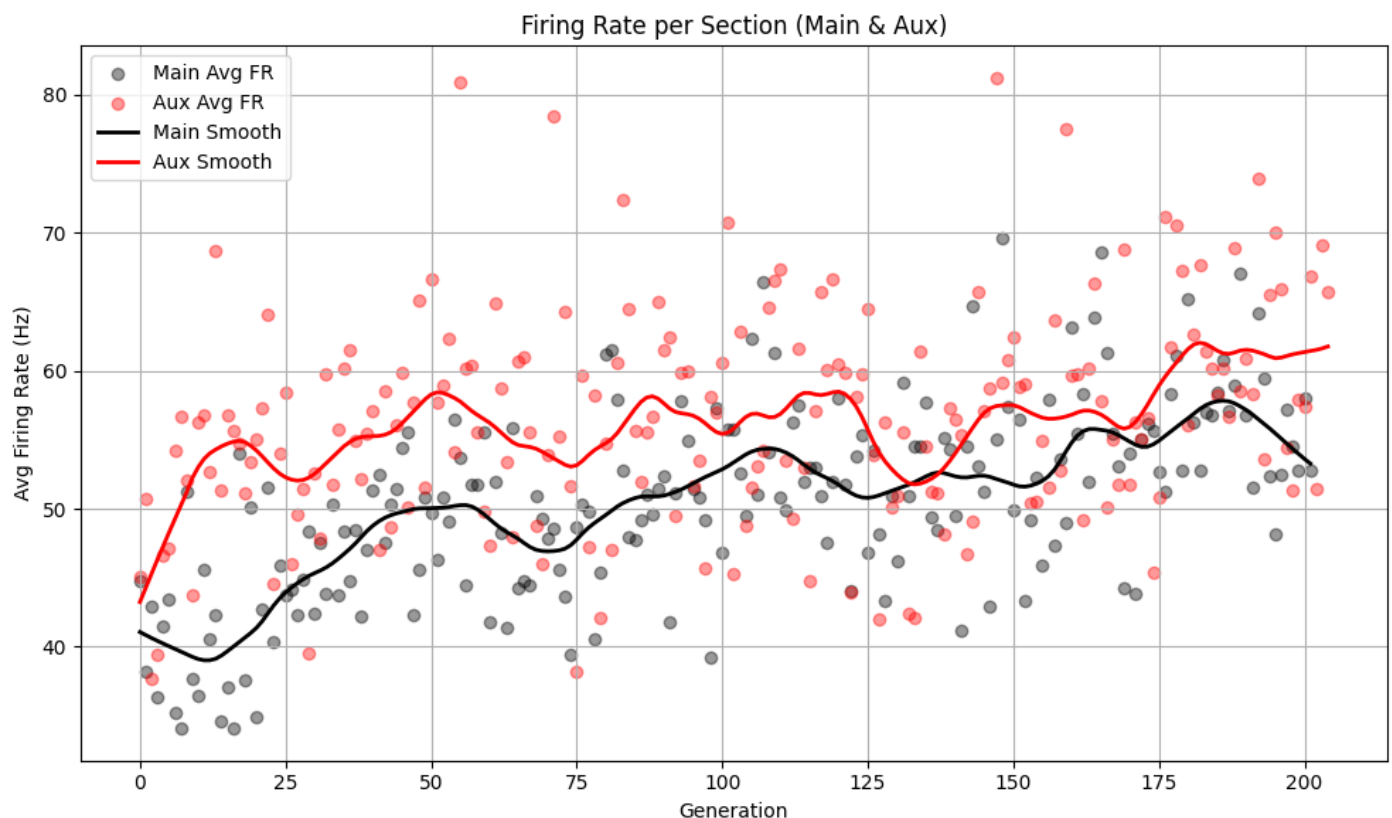
Abstract

This report presents a quantitative and conceptual replication of a closed-loop visual-stimulus paradigm in macaque inferotemporal cortex (ITC). Using trimmed electrophysiological recordings, we extracted per-trial firing rates from both “main” evolution blocks—where a generative algorithm iteratively refined synthetic images based on neural responses—and interleaved “auxiliary” blocks presenting natural images. We implemented a robust analysis pipeline to (1) identify valid main-section blocks (exactly 64 successful trials), (2) compute and compare firing-rate trajectories across trials and sections, and (3) statistically validate the progressive increase in neural responsiveness exclusive to evolution blocks. Our results demonstrate a clear, significant ramp-up of firing rates over generations in the closed-loop main sections, while auxiliary-section responses remain essentially flat, confirming that the genetic-algorithm fitness function successfully targeted and amplified stimulus features driving ITC activity. Finally, we propose alternative fitness criteria—such as phase-locking value or latency shifts—to probe different neuronal response properties and discuss how these would shape the evolved image templates.

Question 1 – Replicating closed-loop experiment

1.1 Results and Interpretations

The hallmark of a successful closed-loop experiment is a systematic increase in neural responsiveness over successive generations of stimuli, and our analysis of the main and auxiliary sections confirms exactly this pattern. First, we extracted firing-rate timecourses for every valid evolution (main) block—i.e., those with exactly 64 successful trials—and compared them to interleaved auxiliary blocks presenting natural images. By fitting a simple linear regression of firing rate versus trial (generation) index within each block, we obtained a distribution of “slopes” quantifying how neural responses changed over time. In the main sections (202 valid blocks), the mean slope was **+0.0114 Hz per trial**, indicating that each successive stimulus, selected by the genetic algorithm, elicited progressively stronger firing. By contrast, the auxiliary sections (205 blocks) showed a mean slope of **−0.136 Hz per trial**, effectively flat or slightly declining over trials. This difference in slope distributions was highly significant (Mann–Whitney U $p = 0.0024$; Welch’s t-test $p = 0.0021$), demonstrating that the ramp-up in firing rate is specific to the evolution phases and not an artifact of general drift or non-specific adaptation.



Visual inspection of the smoothed firing-rate curves further underscores this conclusion: the main-section curve climbs steadily across generations, whereas the auxiliary-section curve remains essentially level. Although auxiliary trials—comprising natural images—have a higher overall mean firing rate (56.4 Hz versus 50.6 Hz for main trials), absolute responsiveness is secondary to the closed-loop paradigm’s goal of **increasing** selectivity. Crucially, only the stimuli chosen by the genetic algorithm in the main blocks drove a

consistent, significant upward trend in neural firing. Together, these quantitative and visual findings replicate the key result of the closed-loop design: that iterative, response-guided stimulus generation can shape and amplify neural activation in inferotemporal cortex.

Question 2 – Fitness function of the Genetic Algorithm

2.1 Alternatives

One natural alternative fitness function is the neuron's **mean-spike latency** within each stimulus window, rather than its overall firing rate. Concretely, for every trial you would measure the latency of *every* spike after image onset in seconds—then compute the trial's average latency. Images that drive spikes more promptly receive higher fitness scores and are preferentially bred into the next generation.

Mean-spike latency emphasizes how quickly the neuron's entire spike train is recruited by the stimulus, capturing both early and sustained activation. It effectively rewards images that touch on features to which the neuron responds with both immediacy and consistency, rather than simply maximizing the total spike count.

$$L_{trial} = \frac{1}{N} \sum_{i=1}^N \left(\frac{t_i - t_{on}}{f_s} \right)$$

- We would expect the final-generation images to highlight the precise combination of contours, textures, or luminance contrasts that engage the neuron's receptive-field subunits with minimal delay.
- Fast-latency stimuli often coincide with strong feedforward drive or optimal alignment with the cell's preferred features, so the resulting images may resemble the neuron's classical preferred stimulus but sharpened to elicit the earliest possible response.

A closed-loop evolution under mean-latency fitness would reveal not only *which* features the neuron encodes (as in firing-rate paradigms) but *how rapidly* it processes those features. Comparing mean-latency trajectories in main versus auxiliary sections would show whether the genetic algorithm is sculpting images that accelerate the neuron's dynamics—a complementary dimension to firing-rate ramp-up and a powerful window into the temporal precision of cortical coding.